

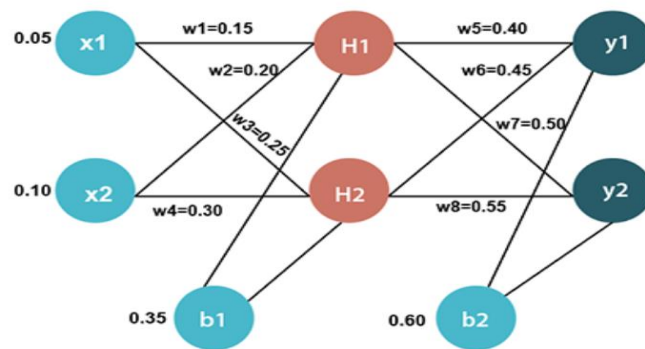
Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.

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|--|-----------|---|
| A) Explain the training procedure of Neural Network. | (BT2/CO1) | 6 |
| B) Explain various components of neural network. | (BT1/CO1) | 6 |
| C) What is perceptron? Compute the output of following neural network. | (BT3/CO1) | 6 |

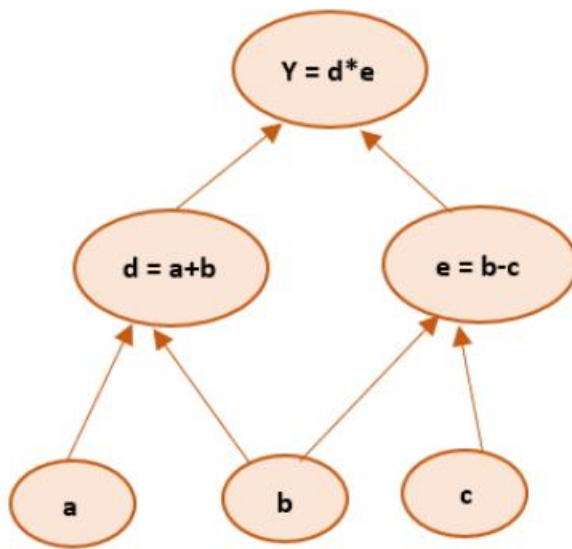


Q.2 Solve Any Two of the following.

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|--|-----------|---|
| A) What is the role of Activation Function? Explain any four Activation function with formula and range. | (BT3/CO2) | 6 |
| B) What is significance of loss function? Explain any four loss function with formula and use. | (BT3/CO2) | 6 |
| C) What is optimizer? Explain Gradient descent algorithm in detail. | (BT3/CO2) | 6 |

Q. 3 Solve Any Two of the following.

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|--|-----------|---|
| A) What is computation graph? Find dY/da , dY/db , dY/dc . | (BT3/CO3) | 6 |
|--|-----------|---|



B) Define backpropagation? Describe basic steps in backpropagation leaning. (BT2/CO3) 6

C) Explain feedforwad Neural Network with suitable example. (BT3/CO1) 6

Q.4 Solve Any Two of the following.

A) Define overfitting and underfittng. Explain various ways (any four) to handle overfitting? (BT3/CO4) 6

B) Define training, testing and validation split? Explain k fold cross validation? (BT2/CO4) 6

C) Describe various steps in model training and evaluation with tensorflow and keras? (BT3/CO4) 6

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Q. 5 Solve Any Two of the following.

A) Differentiate between fuzzy set and Boolean set? Explain union, intersection, difference and complement operation on fuzzy set. (BT3/CO5) 6

B) Explain fuzzification and defuzzification with suitable diagram and example. (BT4/CO5) 6

C) What is Genetic Algorithm? Explain any four application of Genetic Algorithm (BT3/CO5) 6

***** End *****